Multiple Results Chart of Startup Profit Predictive Analysis using 1.Multiple Linear Regression , 2.SVM & 3.Decision Tree Regression

1.Multiple Linear Regression :

Regressor: LinearRegression() (Default Parameters)

R2 Score : **0.9358680970046243** **Predicted Profit Result: 44586.30**

2.SVM (Support Vector Machine) Regression :

Regressor:
Parameters Tweaked are kernel and C - SVR(kernel =“ ” ,C =” ”)

kernel Values are **: "linear","poly","rbf" & "sigmoid"**

c_vales are :**0.025, 0.05, 0.1, 0.5, 1.0, 1.5**

R2 Score :

For C =0.025, R_Square Scores: r_score_linear = 0.9355209342784299 r_score_Poly = -0.05732080221260971 r_score_RBF = -0.05748206464442651 r_score_Sigmoid = -0.057486613037766476	For C =0.5, R_Square Scores: r_score_linear = 0.9230113113764172 r_score_Poly = -0.054182215904108544 r_score_RBF = -0.05740178748055347 r_score_Sigmoid = -0.0574927436944947
For C =0.05, R_Square Scores: **r_score_linear = 0.9390871668294954** r_score_Poly = -0.057155343990719754 r_score_RBF = -0.05747783897509007 r_score_Sigmoid = -0.05748693570044017	For C =1.0, R_Square Scores: r_score_linear = 0.8950779234313312 r_score_Poly = -0.050890117824376135 r_score_RBF = -0.05731730927224388 r_score_Sigmoid = -0.0574991971677592
For C =0.1, R_Square Scores: r_score_linear = 0.9375216515526494 r_score_Poly = -0.056824517369874705 r_score_RBF = -0.057469387821565965 r_score_Sigmoid = -0.05748758102694351	For C =1.5, R_Square Scores: r_score_linear = 0.8985321607927548 r_score_Poly = -0.047061369861496116 r_score_RBF = -0.057232855750551925 r_score_Sigmoid = -0.057505650795273144

Best R2 Score for the parameters are kernel = linear & C =0.05 and it's value is: 0.9390871668294954

Predicted Profit Result of the chosen model is : 44394.75

3.Decision Tree Regression:

Regressor :
Parameters Tweaked are criterion , splitter and max_features

criterion values are : "squared_error", "absolute_error" & "poisson"
splitter values are : "best"& "random"
max_features values are : "sqrt" , "log2", 0.5, 0.6, 0.7, 0.8 & 0.9

R2 Score :

1. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :sqrt. are : 0.49376528621808335
2. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :log2. are : 0.78595656172054
3. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :0.5. are : 0.7637520767864423
4. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :0.6. are : 0.9481841681231585
5. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :0.7. are : 0.756948682694516
6. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :0.8. are : 0.7423000077754102
7. The result of the parametric combinations OF criterion :squared_error splitter :best max_features :0.9. are : 0.7356247083987663
8. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :sqrt. are : 0.4846348792178714
9. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :log2. are : 0.17483647135007752
10. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :0.5. are : 0.6164176379856147
11. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :0.6. are : 0.4382606271202928
12. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :0.7. are : 0.8936504026266254
13. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :0.8. are : 0.9070427610263896
14. The result of the parametric combinations OF criterion :squared_error splitter :random max_features :0.9. are : 0.8919637817846409
15. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :sqrt. are : 0.6865758543325604
16. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :log2. are : 0.7930316926560225
17. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :0.5. are : 0.910639247883286
18. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :0.6. are : 0.6774737572434544
19. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :0.7. are : 0.508593963491398
20. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :0.8. are : 0.9164781322528359
21. The result of the parametric combinations OF criterion :absolute_error splitter :best max_features :0.9. are : 0.9678001862942439
22. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :sqrt. are : -0.24152848626345103
23. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :log2. are : 0.7268601734415707
24. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :0.5. are : -0.4775162236762964
25. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :0.6. are : 0.679226147313505
26. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :0.7. are : 0.6761541108486653
27. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :0.8. are : 0.621517582049308
28. The result of the parametric combinations OF criterion :absolute_error splitter :random max_features :0.9. are : 0.6560336785877622
29. The result of the parametric combinations OF criterion :poisson splitter :best max_features :sqrt. are : 0.5921864421236449
30. The result of the parametric combinations OF criterion :poisson splitter :best max_features :log2. are : -0.156699803909512
31. The result of the parametric combinations OF criterion :poisson splitter :best max_features :0.5. are : 0.741893064383379
32. The result of the parametric combinations OF criterion :poisson splitter :best max_features :0.6. are : 0.7496175062770543
33. The result of the parametric combinations OF criterion :poisson splitter :best max_features :0.7. are : 0.9213283445240934
34. The result of the parametric combinations OF criterion :poisson splitter :best max_features :0.8. are : 0.9271806456008901
35. The result of the parametric combinations OF criterion :poisson splitter :best max_features :0.9. are : 0.9067202900372074
36. The result of the parametric combinations OF criterion :poisson splitter :random max_features :sqrt. are : 0.006862884443456285
37. The result of the parametric combinations OF criterion :poisson splitter :random max_features :log2. are : 0.8720388479999563
38. The result of the parametric combinations OF criterion :poisson splitter :random max_features :0.5. are : 0.6140876448970132
39. The result of the parametric combinations OF criterion :poisson splitter :random max_features :0.6. are : 0.8954843806988316
40. The result of the parametric combinations OF criterion :poisson splitter :random max_features :0.7. are : 0.612430957222329
41. The result of the parametric combinations OF criterion :poisson splitter :random max_features :0.8. are : 0.879522055117929
42. The result of the parametric combinations OF criterion :poisson splitter :random max_features :0.9. are : 0.6926459670740703

The max value among the Result is S.No :21 Which is 0.9678001862942439

Best R2 Score for the parameters are criterion =absolute_error , splitter = best & max_features : 0.9 and it's value is: 0.9678001862942439

Predicted Profit Result of the chosen model is : 49490.75

4.Random Forest Regression:

Regressor :
Parameters Tweaked are criterion , n_estimator and max_features

criterion values are : "squared_error", "absolute_error" & "poisson"
n_estimator values are : 10 , 20 , 50 & 100
max_features values are : 0.6, 0.7, 0.8 & 0.9

R2 Score :

1. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:10 , Max_features:0.6 . are: 0.9141903451204532
2. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:10 , Max_features:0.7 . are: 0.8659672515421866
3. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:10 , Max_features:0.8 . are: 0.8920286614564704
4. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:10 , Max_features:0.9 . are: 0.9597240918539941
5. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:20 , Max_features:0.6 . are: 0.8848799855796751
6. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:20 , Max_features:0.7 . are: 0.9035759983585014
7. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:20 , Max_features:0.8 . are: 0.9171393889800197
8. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:20 , Max_features:0.9 . are: 0.9324696369864374
9. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:50 , Max_features:0.6 . are: 0.8681566098503424
10. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:50 , Max_features:0.7 . are: 0.9020552068737926
11. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:50 , Max_features:0.8 . are: 0.9261404266750666
12. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:50 , Max_features:0.9 . are: 0.8950381701388752
13. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:100 , Max_features:0.6 . are: 0.8957777708772842
14. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:100 , Max_features:0.7 . are: 0.8794356980042868
15. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:100 , Max_features:0.8 . are: 0.9010358986593804
16. The result of the parametric combinations of Criterion:squared_error , Estimator_Count:100 , Max_features:0.9 . are: 0.9098605207296884
17. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:10 , Max_features:0.6 . are: 0.8396536534042713
18. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:10 , Max_features:0.7 . are: 0.9217431589600324
19. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:10 , Max_features:0.8 . are: 0.9018932125688457
20. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:10 , Max_features:0.9 . are: 0.9524304835581942
21. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:20 , Max_features:0.6 . are: 0.8983850737478244
22. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:20 , Max_features:0.7 . are: 0.8906599749134201
23. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:20 , Max_features:0.8 . are: 0.9139670175230609
24. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:20 , Max_features:0.9 . are: 0.9230641076052533
25. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:50 , Max_features:0.6 . are: 0.9054131270488059
26. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:50 , Max_features:0.7 . are: 0.8969797681559091
27. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:50 , Max_features:0.8 . are: 0.9149260287249955
28. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:50 , Max_features:0.9 . are: 0.9304495592458002
29. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:100 , Max_features:0.6 . are: 0.8974719221301002
30. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:100 , Max_features:0.7 . are: 0.8991227505282162
31. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:100 , Max_features:0.8 . are: 0.9160065090655556
32. The result of the parametric combinations of Criterion:absolute_error , Estimator_Count:100 , Max_features:0.9 . are: 0.9416987571326318
33. The result of the parametric combinations of Criterion:poisson , Estimator_Count:10 , Max_features:0.6 . are: 0.844493758481176
34. The result of the parametric combinations of Criterion:poisson , Estimator_Count:10 , Max_features:0.7 . are: 0.9054206761374938
35. The result of the parametric combinations of Criterion:poisson , Estimator_Count:10 , Max_features:0.8 . are: 0.9149115973913684
36. The result of the parametric combinations of Criterion:poisson , Estimator_Count:10 , Max_features:0.9 . are: 0.925663440690179
37. The result of the parametric combinations of Criterion:poisson , Estimator_Count:20 , Max_features:0.6 . are: 0.8180073153526913
38. The result of the parametric combinations of Criterion:poisson , Estimator_Count:20 , Max_features:0.7 . are: 0.891283812013811
39. The result of the parametric combinations of Criterion:poisson , Estimator_Count:20 , Max_features:0.8 . are: 0.898602114792432
40. The result of the parametric combinations of Criterion:poisson , Estimator_Count:20 , Max_features:0.9 . are: 0.8632261488079613
41. The result of the parametric combinations of Criterion:poisson , Estimator_Count:50 , Max_features:0.6 . are: 0.881710651255076
42. The result of the parametric combinations of Criterion:poisson , Estimator_Count:50 , Max_features:0.7 . are: 0.8565374895514182
43. The result of the parametric combinations of Criterion:poisson , Estimator_Count:50 , Max_features:0.8 . are: 0.9371722541716713
44. The result of the parametric combinations of Criterion:poisson , Estimator_Count:50 , Max_features:0.9 . are: 0.9391614644994645
45. The result of the parametric combinations of Criterion:poisson , Estimator_Count:100 , Max_features:0.6 . are: 0.8623634580484125
46. The result of the parametric combinations of Criterion:poisson , Estimator_Count:100 , Max_features:0.7 . are: 0.8967329927332053
47. The result of the parametric combinations of Criterion:poisson , Estimator_Count:100 , Max_features:0.8 . are: 0.9167727382083173
48. The result of the parametric combinations of Criterion:poisson , Estimator_Count:100 , Max_features:0.9 . are: 0.9309533600987212

the max value among the Result is :4
0.9597240918539941 RandomForestRegressor(max_features=0.9, n_estimators=10)

Predicted Result of the chosen model is Profit of : 53889.2790 (Which is the best Resultof all Models)